# Note 2.1 Sets

## Introduction

In this section, we study the set which is the fundamental discrete structure on which all other discrete structures are built.

- **Definition of Set(Intuitive Definition)**
  A set is an unordered collection of objects, called elements or members of the set. A set is said to contain its elements. We write $a \in A$ to denote that $a$ is an element of the set $A$. The notation $a \notin A$ denotes that $a$ is not an element of the set $A$.
  It is common for sets to be denoted using uppercase letters. Lowercase letters are usually used to denote elements of sets.
- **Ways to Describe a Set**
  - **Roster Method**
    A notation where all members of the set are listed between braces.
  - **Set Builder Notation**
    We characterize all those elements in the set by stating the property or properties they must have to be members.
- **Some Special Sets**

$$
\begin{aligned}
&\mathbf{N} = \{0, 1, 2, \cdots\}, \text{the set of } \textbf{natural numbers}\\
&\mathbf{Z} = \{\cdots, -1, 0, 1, \cdots\}, \text{the set of } \textbf{integers}\\
&\mathbf{Z}^+ = \{1, 2, \cdots\}, \text{the set of } \textbf{positive numbers}\\
&\mathbf{Q} = \left\{\frac{p}{q} | p \in \mathbf{Z}, q \in \mathbf{Z}, q \neq 0\right\}, \text{the set of } \textbf{rational numbers}\\
&\mathbf{R}, \text{the set of } \textbf{real numbers}\\
&\mathbf{R}^+, \text{the set of } \textbf{positive real numbers}\\
&\mathbf{C}, \text{the set of } \textbf{complex numbers}
\end{aligned}
$$

- **Intervals**
  closed interval and open interval
- **Datatype in Computer Science**
  a datatype or type is the name of a set, together with a set of operations that can be performed on objects from that set.
- **Equal Sets**
  Two sets are equal if and only if they have the same elements. Therefore, if $A$ and $B$ are sets, then $A$ and $B$ are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$. We write $A = B$ if $A$ and $B$ are equal sets.
- **Empty Set(Null Set) and Singleton Set**
  There is a special set that has no elements. This set is called the empty set, or null set, and

is denoted by $\emptyset$. It can be denoted by $\{\ \}$ or a set of elements with certain properties turns out to be the null set.

- **Naive Set Theory**
  Note that the term object has been used in the definition of a set without specifying what an object is. And the description of a set as a collection of objects based on the intuitive notation of a object. So it might lead to paradoxes that there exists a set containing the objects with the same property no matter the property is. And the theory to use intuitive notion is Naive Set theory.

## Venn Diagrams

In Venn diagrams the universal set $U$, which contains all the objects under consideration, is represented by a rectangle. And circles or other geometrical figures are used to represent sets inside the rectangle. Sometimes points are used to represent the particular elements of the set.

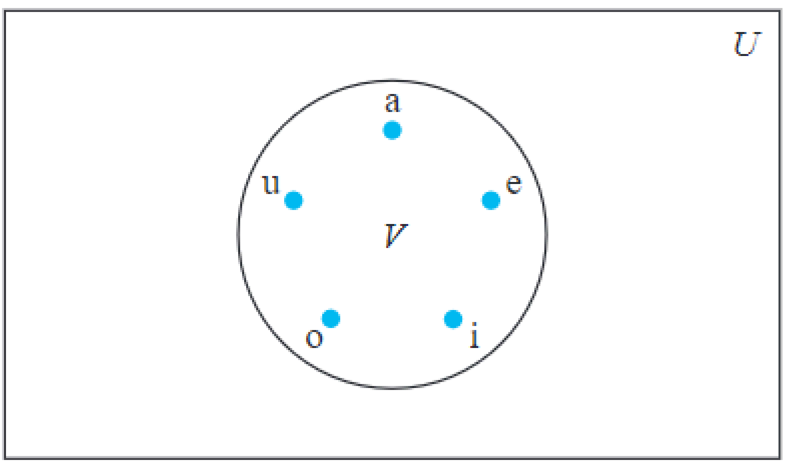


**FIGURE 1 Venn Diagram for the Set of Vowels.**

## Subsets

- **Definition of Subsets**
  The set $A$ is a subset of $B$ if and only if every element of $A$ is also an element of $B$. We use the notation $A \subseteq B$ to indicate that $A$ is a subset of the set $B$.
  And $A \subseteq B$ is equivalent to

$$\forall x(x \in A \rightarrow x \in B)$$

  is true.
  And if $x$ belongs to $A$ then $x$ also belongs to $B$, then we can prove $A$ is a subset of $B$. If find

a single $x \in A$ such that $x \notin B$, then we can prove $A$ is not a subset of $B$.
To show two sets are equal, we can prove that $A \subseteq B \text{ and } B \subseteq A$.

- **Theorem 1**
  For every set $S$,
  - $\emptyset \subseteq S$
  - $S \subseteq S$
- **Proper Subset**
  If a set $A$ is a subset of $B$ and $A \neq B$, we write $A \subset B$ and say $A$ is a proper subset of $B$.
  That is equivalent to

$$\forall x(x \in A \to x \in B) \land \exists x(x \in B \land x \notin A)$$

  is true.

# The Size of a Set

- **Definition of the size of set**
  Let $S$ be a set. If there are exactly $n$ distinct elements in $S$ where $n$ is a nonnegative integer, we say that $S$ is a finite set and that $n$ is the cardinality of $S$. The cardinality of $S$ is denoted by $|S|$.
- **Definition of Infinite Set**
  A set is said to be infinite if it is not finite.

# Power Sets

- **Definition of Power Sets**
  Given a set $S$, the power set of $S$ is the set of all subsets of the set $S$. And it is denoted by $\mathcal{P}(S)$.
- **Remarks**
  If a set has $n$ elements, then its power set has $2^n$ elements.

# Cartesian Products

- **Definition of ordered $n$-tuples**
  The ordered $n$-tuple $(a_1, a_2, \cdots, a_n)$ is the ordered collection that corresponding element is fixed on the corresponding position.
  We say that two ordered $n$-tuples are equal if and only if each corresponding pair of their elements is equal.
  In particular, the ordered $2$-tuples are called ordered pairs.
- **Definition of Cartesian Products**
  Let $A$ and $B$ be sets. The Cartesian product of $A$ and $B$, denoted by $A \times B$, is the set of all

ordered pairs $(a,b)$, where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a,b) \mid a \in A \land b \in B\}$$

Note that Cartesian product does not satisfy the commutative law and associative law generally, that is, $A \times B$ not equals to $B \times A$ unless $A = \emptyset$ or $B = \emptyset$ or $A = B$ and $A \times B \times C \neq (A \times B) \times C \neq A \times (B \times C)$.
For the sets $A_1, A_2, \cdots, A_n$, its Cartesian product is the set of ordered $n$-tuples $(a_1, a_2, \cdots, a_n)$ where $a_i$ belongs to $A_i$ for $i = 1, 2, \cdots, n$. That is,

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \cdots, a_n) | a_i \in A_i, i = 1, 2, \cdots, n\}$$

And $A^n$ represents the Cartesian product of $n$ $A$.

- **Definition of Relation**
  A subset $R$ of the Cartesian product $A \times B$ is called a relation from the set $A$ to the set $B$. A relation from a set $A$ to itself is called a relation on $A$.

# Using Set Notation with Quantifiers

Sometimes we can restrict the domain of a quantified statement explicitly by making use of a particular notation. For example,

$$\forall x \in S(P(x))$$

# Truth Sets and Quantifiers

Given a predicate $P$ and a domain $D$, we define the truth set of $P$ to be the set elements $x$ in $D$ for which $P(x)$ is true and denoted by $\{x \in D | P(x)\}$.